Forecasting Euro-Area Growth, Inflation and Interest Rates: A Small DSGE Model versus a Bayesian VAR

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Abstract

This study compares a small New Keynesian dynamic stochastic general equilibrium (DSGE) model, a Bayesian vector autoregression (BVAR) and simple benchmarks for forecasting euro-area output growth, inflation and the short-term interest rate. Quarterly observations are constructed from ECB real GDP, HICP and three-month Euribor series. Both models are estimated on 1999–2019 data and evaluated over 2020–2025 at one- and four-quarter horizons. Parameters and posterior weights remain fixed, while states and observed lags are updated. The DSGE model uses a Kalman-filter likelihood and Markov chain Monte Carlo; the BVAR uses a conjugate normal–inverse-Wishart prior. The DSGE model has lower RMSE, interval score and CRPS than the BVAR in five of six variable–horizon comparisons. However, benchmarks perform best for one-quarter-ahead growth and interest rates, and rankings change across subperiods. Validation reveals discrepancies in extreme output contractions, inflation dynamics and the interest-rate distribution. Relative performance therefore varies across variables, horizons and economic episodes. The project provides an applied Python workflow combining structural modelling, Bayesian estimation, validation and forecast evaluation.

1 Introduction

1.1 Motivation and research question

Macroeconomic forecasts inform assessments of the economic outlook and its risks. Forecasting growth, inflation and interest rates requires a framework for their joint dynamics, but modelling approaches impose different restrictions on these relationships. Structural models connect variables through assumptions about economic behaviour, while statistical models allow more flexible lagged relationships. Comparing their forecasts assesses the predictive performance of these alternative specifications.

This project compares a small New Keynesian dynamic stochastic general equilibrium (DSGE) model with a Bayesian vector autoregression (BVAR). The DSGE model combines consumption habits, Calvo pricing, inflation indexation and an interest-rate feedback rule, providing explicit mechanisms for shock transmission [Calvo, 1983, Christiano et al., 2005, Taylor, 1993]. The BVAR uses Bayesian shrinkage to regularise the observed variables' lagged relationships [Karlsson, 2013]. Both models generate predictive distributions, while the DSGE model also assigns economic interpretations to its disturbances.

The research question is:

> How does a small estimated New Keynesian DSGE model compare with a BVAR and simple benchmarks when forecasting euro-area output growth, inflation and the short-term interest rate at one- and four-quarter horizons?

The analysis uses ECB real GDP, HICP and three-month Euribor series to construct quarterly growth, annualised quarterly inflation and a quarterly average interest rate [European Central Bank, n.d.b,n,n]. Estimation covers 1999Q1–2019Q4, followed by forecast evaluation over 2020–2025. The pandemic and subsequent inflation and interest-rate adjustments make this a demanding test of pre-2020 estimates. Subperiod comparisons assess how these episodes affect the results.

The comparison considers both point forecasts and predictive distributions. It also includes model validation, since forecast accuracy alone cannot establish the empirical adequacy or structural identification of the DSGE model.

1.2 Scope and contribution

The contribution is an applied Python implementation connecting data preparation, structural model solution, Bayesian estimation, validation and forecast evaluation. The DSGE specification uses established theoretical building blocks rather than proposing a new economic model. Its linear solution and state-space representation support Bayesian estimation and conditional impulse-response analysis [Klein, 2000, An and Schorfheide, 2007]. Computational diagnostics, posterior predictive checks and calibration sensitivity complement a common

forecasting experiment for the DSGE model, BVAR and benchmarks. Evaluation covers point forecast errors and predictive calibration, including proper scoring rules [Gneiting and Raftery, 2007].

The scope is limited to a small linear DSGE model, a BVAR(2) with predetermined prior settings and a short evaluation period. The exercise uses revised data rather than historical real-time vintages, and DSGE inference is conditional on fixed calibration. The findings therefore concern these specifications and the chosen forecasting protocol.

The report proceeds from data and model specification to estimation, validation and forecast evaluation, followed by results and discussion. The appendices provide technical derivations, additional diagnostics and results, and a reproducibility guide.

2 Data

2.1 Sources and variable definitions

The empirical analysis uses three series from the European Central Bank (ECB) Data Portal: real gross domestic product (GDP), the all-items Harmonised Index of Consumer Prices (HICP), and the three-month Euribor rate [European Central Bank, n.d.b,n,n]. These series provide observations on real activity, consumer price inflation, and short-term nominal interest rates for both the DSGE and BVAR models. Table 1 summarises their measurement conventions.

Table 1: Source series and measurement conventions

Series	Frequency	Source unit	Geographic coverage
Real GDP	Quarterly	Chain-linked volume, EUR millions; reference year 2015	Fixed euro area composition of 20 countries (EA20)
All-items HICP	Monthly	Price index, 2025 = 100	Euro area, changing composition
Three-month Euribor	Monthly	Percent per annum; monthly average	Euro-denominated money market; source area code U2

The exact ECB series identifiers are:

- Real GDP: `MNA.Q.Y.I9.W2.S1.S1.B.B1GQ._Z._Z._Z.EUR.LR.N`.
- All-items HICP: `HICP.M.U2.Y.000000.4F0.INX`.
- Three-month Euribor: `FM.M.U2.EUR.RT.MM.EURIBOR3MD_.HSTA`.

GDP is supplied in calendar- and seasonally-adjusted form, while the HICP series is working-day- and seasonally-adjusted. The GDP level and HICP index are used to construct growth and inflation rates rather than entering the models

as raw levels. Euribor is already quoted as an annual interest rate and is retained in percent per annum. Its source metadata identifies Refinitiv as the provider and describes the monthly observations as averages over the period.

The series definitions and index bases reported here correspond to the archived CSV files used in the analysis. Subsequent downloads may contain revised observations or updated metadata.

2.2 Transformations and sample construction

The raw GDP series contains 112 quarterly observations from 1998Q1 to 2025Q4. Each monthly series contains 336 observations from January 1998 to December 2025. Validation of the archived files finds no missing values, duplicate dates, or gaps in the required calendars.

Let Y_q denote real GDP in quarter q, P_m the monthly HICP index, and R_m the monthly Euribor rate in percent per annum. The monthly series are converted to quarterly observations using unweighted arithmetic averages:

$$\overline{P}_q = \frac{1}{3}\sum_{m\in q} P_m, \qquad i_q^{\text{obs}} = \frac{1}{3}\sum_{m\in q} R_m. \tag{1}$$

All three monthly observations are required for a quarter to be retained. Each month receives equal weight; the aggregation is not weighted by the number of calendar days or market observations.

The observable growth and inflation rates are defined as

$$g_q^{\text{obs}} = 100\left(\log Y_q - \log Y_{q-1}\right), \tag{2}$$

$$\pi_q^{\text{obs}} = 400\left(\log \overline{P}_q - \log \overline{P}_{q-1}\right). \tag{3}$$

GDP growth is measured in log percent per quarter, whereas inflation is measured in annualised log percent. Inflation is therefore calculated from changes in the quarterly average price index. It is neither a year-on-year inflation rate nor an average of monthly inflation rates.

The factor of four annualises the quarterly log price change. Euribor is not multiplied or divided by four, since its source observations are already expressed in annual terms. Multiplicative changes in the units of GDP or the base of the HICP index cancel in the log differences.

The common model sample consists of 108 quarters from 1999Q1 to 2025Q4. The GDP and quarterly HICP levels for 1998Q4 provide the lagged observations needed to calculate the first growth and inflation rates. Observations before 1999 are not included in the model sample.

The estimation sample contains 84 quarters from 1999Q1 to 2019Q4. The remaining 24 quarters, from 2020Q1 to 2025Q4, form the forecast evaluation sample. This chronological split is used for both models.

No observations are interpolated, winsorised, or removed as outliers. No additional seasonal adjustment or trend filter is applied, and the prepared observables are not demeaned or standardised. The BVAR subsequently uses a

model-specific standardisation based exclusively on the estimation sample, as described in the forecasting methodology.

2.3 Descriptive statistics and data limitations

Figure 1 displays the transformed series. Table 2 reports descriptive statistics separately for the estimation and test samples. Test-sample statistics are presented to characterise the evaluation period and are not used to select model settings.

Table 2: Descriptive statistics of the model observables

Variable	N	Mean	SD	Minimum	Maximum
Estimation sample: 1999Q1–2019Q4					
GDP growth	84	0.358	0.606	−3.094	1.287
HICP inflation	84	1.677	1.270	−2.224	4.568
Three-month Euribor	84	1.747	1.749	−0.403	5.024
Test sample: 2020Q1–2025Q4					
GDP growth	24	0.269	3.487	−11.818	10.919
HICP inflation	24	3.463	3.266	−1.469	10.664
Three-month Euribor	24	1.425	1.816	−0.566	3.958

Notes: GDP growth is measured in log percent per quarter, inflation in annualised log percent, and Euribor in percent per annum. SD denotes the sample standard deviation, calculated with denominator $N - 1$. Source: ECB Data Portal and author calculations.

The test period contains substantially larger movements in output and prices than the estimation period. The standard deviation of quarterly GDP growth rises from 0.606 to 3.487, with a sharp contraction and rebound in 2020. Mean inflation increases from 1.677 to 3.463, while its standard deviation rises from 1.270 to 3.266. The interest-rate series spans both negative rates and a subsequent increase in rates. These features make the test period a demanding assessment of models estimated on the earlier sample.

Three measurement limitations affect the interpretation of the results. First, the data constitute a single revised snapshot rather than historical real-time vintages. Revisions to GDP and seasonal adjustment may incorporate information that was unavailable at the historical forecast origin. Publication delays are not reproduced. The exercise is therefore a historical forecast evaluation using revised data, rather than a real-time forecasting experiment.

Second, GDP refers to a fixed EA20 composition, whereas HICP refers to the changing composition of the euro area. The series consequently do not represent exactly the same geographic aggregate throughout the sample. This mismatch is retained and documented rather than treated as a fully harmonised dataset.

Third, three-month Euribor is a market interest rate rather than an ECB policy rate or a risk-free short rate. Its use as the interest-rate observable is an approximation: movements can also reflect market expectations and credit or liquidity premia. This qualification matters when interpreting the DSGE

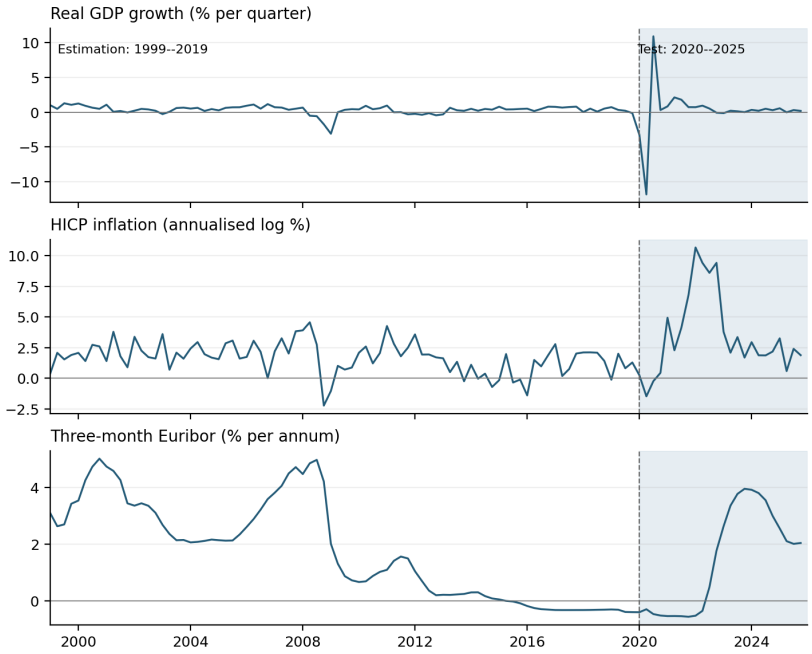


Figure 1: Quarterly model observables, 1999Q1–2025Q4. The shaded region identifies the test sample beginning in 2020Q1. Inflation is annualised quarter-on-quarter HICP inflation, not year-on-year inflation. Source: ECB Data Portal and author calculations.

model's monetary-policy equation and shocks.

The archived HICP metadata additionally records methodological changes in national components. These comments are retained in the data audit, but no separate adjustments for such changes are imposed in this project.

3 The DSGE Model

3.1 Economic environment and assumptions

The model is a small closed-economy New Keynesian model with a representative household, monopolistically competitive intermediate producers, a competitive final-goods sector and a monetary authority. External consumption habits, Calvo price adjustment, inflation indexation and interest-rate smoothing generate persistence. These mechanisms draw on established New Keynesian modelling approaches [Calvo, 1983, Christiano et al., 2005]; the specification used here is a deliberately small implementation.

There is no capital accumulation, government expenditure, external sector or banking sector. Aggregate consumption equals output, and productivity grows deterministically:

$$C_t = Y_t, \tag{4}$$

$$A_t = G A_{t-1}. \tag{5}$$

The model is approximated to first order around a balanced-growth equilibrium with constant labour supply. It includes neither stochastic trend productivity nor an effective lower bound on interest rates.

Households choose consumption, labour and nominal bond holdings. Consumption utility depends on $C_t - h_0 C^a_{t-1}$, where lagged aggregate consumption C^a_{t-1} is taken as given, making habits external. Intermediate producers use labour and face a probability $1-\theta$ of resetting their price each quarter. Non-reset prices are indexed to steady-state inflation and, with coefficient ι, to lagged inflation deviations. Monetary policy follows a smoothed interest-rate feedback rule.

The underlying optimisation problems and the derivation of the linear equilibrium conditions are presented in Appendix A.

3.2 Equilibrium equations and shocks

Let y_t denote the log deviation of detrended output from steady state, and let π_t and r_t denote quarterly log deviations of gross inflation and the nominal interest rate:

$$y_t = \log\left(\frac{Y_t/A_t}{\overline{\widetilde{Y}}}\right), \tag{6}$$

$$\pi_t = \log\left(\frac{\Pi_t}{\overline{\Pi}}\right), \qquad r_t = \log\left(\frac{R_t}{\overline{R}}\right), \tag{7}$$

where $\Pi_t = P_t/P_{t-1}$. The variable y_t measures output relative to a deterministic trend, rather than an output gap relative to flexible-price output.

The coefficients entering the stationary system are

$$h = \frac{h_0}{G}, \qquad a = \frac{\sigma}{1-h}, \qquad B = \beta G^{1-\sigma}, \qquad \kappa = \frac{(1-\theta)(1-B\theta)}{\theta}.$$

Here β is the household discount factor, σ is consumption-utility curvature and φ is the inverse Frisch elasticity. Admissible parameter combinations satisfy $0 \leq h < 1$ and $0 < B < 1$.

The reduced Euler condition is

$$a(1+h)y_t = a\mathbb{E}_t y_{t+1} + ahy_{t-1} - r_t + \mathbb{E}_t \pi_{t+1} + (1-\rho_b)b_t. \tag{8}$$

Real marginal cost, expressed as a log deviation, is

$$mc_t = (\varphi + a)y_t - ahy_{t-1} - b_t. \tag{9}$$

External habits introduce lagged consumption into both relations. Their derivation is given in Appendix A.2.

The indexed Phillips curve is

$$\pi_t = \frac{B\mathbb{E}_t \pi_{t+1} + \iota \pi_{t-1} + \kappa mc_t}{1 + B\iota} + u_t. \tag{10}$$

Full indexation to steady-state inflation permits a symmetric steady-state relative-price distribution; price dispersion does not enter the first-order system. The cost-push disturbance u_t enters the normalised inflation equation with coefficient one. It is not separately identified as an energy-price or other specific supply shock.

Monetary policy follows a feedback rule with smoothing, drawing on the policy-rule literature [Taylor, 1993]:

$$r_t = \rho_r r_{t-1} + (1-\rho_r)(\phi_\pi \pi_t + \phi_y y_t) + m_t. \tag{11}$$

The coefficients ϕ_π and ϕ_y govern responses to inflation and detrended output. The monetary disturbance enters outside the smoothing term.

The three shock processes are

$$b_t = \rho_b b_{t-1} + s_b \varepsilon_{b,t}, \tag{12}$$

$$u_t = \rho_u u_{t-1} + s_u \varepsilon_{u,t}, \tag{13}$$

$$m_t = s_m \varepsilon_{m,t}, \tag{14}$$

with mutually independent innovations

$$\boldsymbol{\varepsilon}_t = (\varepsilon_{b,t}, \varepsilon_{u,t}, \varepsilon_{m,t})' \overset{\text{iid}}{\sim} \mathcal{N}(0, I_3).$$

The preference and cost-push disturbances are persistent, whereas the monetary disturbance is serially independent. Innovation standard deviations are positive.

The preference disturbance affects both expenditure through the Euler condition and marginal cost through the direct term $-b_t$. Its responses therefore require a more specific interpretation than a generic aggregate-demand shock.

3.3 Solution and observation equations

The state and decision vectors are

$$s_t = (y_{t-1}, \pi_{t-1}, r_{t-1}, b_t, u_t, m_t)', \tag{15}$$

$$d_t = (y_t, \pi_t, r_t)'. \tag{16}$$

Current shocks are observed by economic agents before contemporaneous decisions are determined. The resulting linear rational-expectations system is solved using an ordered generalised Schur decomposition [Klein, 2000]. Parameter combinations are retained only when they imply a unique stable state mapping and satisfy the numerical equilibrium checks.

The solution has the form

$$d_t = H s_t, \tag{17}$$

$$s_{t+1} = F s_t + J \varepsilon_{t+1}. \tag{18}$$

The equilibrium matrices, construction of H, F and J, and solution checks are documented in Appendix A.4.

The model is connected to the data through

$$g_t^{\text{obs}} = \overline{g} + 100(y_t - y_{t-1}), \tag{19}$$

$$\pi_t^{\text{obs}} = \overline{\pi} + 400\pi_t, \tag{20}$$

$$i_t^{\text{obs}} \approx \overline{i} + 400 r_t. \tag{21}$$

Here $\overline{g} = 100 \log G$ and $\overline{\pi} = 400 \log \overline{\Pi}$. The quarterly average HICP index is treated as the counterpart of the model's quarterly price level. The interest-rate equation approximates Euribor's annual percentage quote by an annualised log nominal rate, with no separate market-rate spread or measurement-error process.

The steady-state Euler condition restricts the long-run rate:

$$\overline{i} = \overline{\pi} + 400(\sigma \log G - \log \beta). \tag{22}$$

Thus, the interest-rate intercept is determined by preferences, trend growth and steady-state inflation; it is not independently estimated.

Stacking the observables gives

$$z_t = (g_t^{\text{obs}}, \pi_t^{\text{obs}}, i_t^{\text{obs}})' = \mu_z + C s_t. \tag{23}$$

The observation matrix is reported in Appendix A.5. This state-space representation provides the basis for the likelihood and estimation procedure described in Section 4.

4 Bayesian Estimation and Model Validation

4.1 Estimated parameters, calibration and priors

The model is estimated using 84 quarterly observations from 1999Q1 to 2019Q4. Bayesian inference combines the state-space likelihood with parameter priors,

following the framework reviewed by An and Schorfheide [2007]. The estimated parameter vector is

$$\boldsymbol{\psi} = (h_0, \iota, \rho_r, \phi_\pi, \rho_b, \rho_u, s_b, s_u, s_m, \overline{g}, \overline{\pi})'. \tag{24}$$

Five structural coefficients are fixed:

$$\beta = 0.999, \quad \sigma = 1, \quad \varphi = 1, \quad \theta = 0.75, \quad \phi_y = 0.125. \tag{25}$$

Posterior uncertainty is conditional on this calibration. The effective habit coefficient is calculated as $h = h_0/\exp(\overline{g}/100)$, and the long-run interest rate follows Equation (22).

Beta priors are used for habit, indexation and persistence parameters; a shifted Gamma prior for ϕ_π; lognormal priors for shock standard deviations; and Normal priors for trend growth and inflation. Each distribution is truncated and normalised over its specified bounds. The complete specification is reported in Table 10 in Appendix B.1.

Let $\mathcal{Z}_T = \{z_1, \ldots, z_T\}$ denote the estimation data and $\mathcal{A}$ the set of parameter combinations yielding an admissible, unique stable solution. The posterior satisfies

$$p(\boldsymbol{\psi} \mid \mathcal{Z}_T) \propto L(\mathcal{Z}_T \mid \boldsymbol{\psi}) \prod_{j=1}^{11} p_j(\psi_j) \mathbb{I}\{\boldsymbol{\psi} \in \mathcal{A}\}. \tag{26}$$

The univariate priors are independent before imposing the joint admissibility restriction. The support requires $\phi_\pi > 1$, so a posterior estimate above one is not an independent empirical test of the Taylor principle.

4.2 Likelihood and posterior computation

The Gaussian likelihood is evaluated using a Kalman filter [Kalman, 1960]. For each parameter vector, the model is solved again and its state-space matrices are updated. The filter starts from the stationary distribution

$$s_1 \sim \mathcal{N}(0, P_\infty), \qquad P_\infty = F P_\infty F' + J J'. \tag{27}$$

Writing a_t and P_t^- for the predicted state mean and covariance conditional on observations through $t-1$, the innovations are

$$v_t = z_t - \mu_z - C a_t, \tag{28}$$

$$S_t = C P_t^- C'. \tag{29}$$

With three observables, the log likelihood is

$$\log L = -\frac{1}{2} \sum_{t=1}^{T} \left[3 \log(2\pi) + \log |S_t| + v_t' S_t^{-1} v_t \right]. \tag{30}$$

Cholesky factorisation evaluates the determinant and quadratic form. No measurement noise or covariance jitter is added; invalid parameter evaluations are rejected.

Estimation first obtains a maximum a posteriori (MAP) candidate using L-BFGS-B in transformed coordinates. The two starting points in the final run converge to the same reported objective value. Four random-walk Metropolis chains are then initialised at dispersed points near this candidate [Hastings, 1970]. Each chain uses 5,000 warmup iterations and 20,000 retained draws, producing 80,000 posterior draws without thinning. Proposal adaptation is confined to warmup.

MAP optimisation targets the density in the original parameter coordinates. Sampling uses bounded-parameter transformations and includes the corresponding Jacobian. The filter recursions and sampling details are documented in Appendix B.2, with numerical checks in Appendix B.4 and run settings in Appendix D.

4.3 Convergence and posterior estimates

Convergence is assessed using rank-normalised split and folded $\widehat{R}$, together with bulk and tail effective sample sizes (ESS) [Vehtari et al., 2021]. All parameters satisfy the adopted criteria of $\widehat{R} \leq 1.01$ and both ESS measures at least 400. The maximum $\widehat{R}$ is 1.0047, minimum bulk ESS is 1,103 and minimum tail ESS is 918. Parameter-level diagnostics and trace plots are reported in Appendix B.3. These checks support sampling convergence, rather than establishing empirical model adequacy or structural identification.

Table 3: Posterior estimates of the DSGE parameters

Parameter	Mean	SD	5th percentile	95th percentile
h_0	0.859	0.037	0.794	0.914
ι	0.070	0.056	0.008	0.180
ρ_r	0.895	0.020	0.861	0.926
ϕ_π	1.722	0.347	1.251	2.367
ρ_b	0.750	0.051	0.657	0.824
ρ_u	0.899	0.044	0.819	0.960
$10^3 s_b$	49.638	10.990	34.824	69.598
$10^3 s_u$	1.247	0.235	0.856	1.631
$10^3 s_m$	0.885	0.078	0.766	1.022
$\overline{g}$	0.305	0.033	0.248	0.354
$\overline{\pi}$	1.507	0.524	0.705	2.426

Notes: Percentiles define equal-tailed 90% credible intervals. Shock scales are multiplied by 10^3 for presentation. Trend growth is in log percent per quarter and trend inflation in annualised log percent. Estimates are conditional on the fixed calibration, prior supports and model specification.

Table 3 indicates substantial consumption habits and interest-rate smoothing, limited inflation indexation and persistent cost-push disturbances. The inflation-response coefficient remains comparatively uncertain within its imposed support.

4.4 Posterior predictive checks

Model adequacy is examined using 1,000 posterior predictive replications of 84 quarters. Each replication draws a parameter vector from the posterior, an initial state from its stationary distribution and new independent innovations, following the model-checking approach discussed by An and Schorfheide [2007].

Table 4 compares observed training-sample statistics with the central 90% replication intervals. These are descriptive checks, not formal acceptance tests. Simulation details and complementary figures are provided in Appendix C.3.

Table 4: Selected posterior predictive checks

Statistic	Observed	Replication 5%	Replication 95%
GDP growth: mean	0.358	0.208	0.397
GDP growth: SD	0.606	0.551	0.886
GDP growth: autocorrelation	0.588	0.430	0.716
GDP growth: minimum	−3.094	−1.982	−0.881
Inflation: mean	1.677	0.197	2.934
Inflation: SD	1.270	1.355	2.329
Inflation: autocorrelation	0.387	0.522	0.804
Euribor: mean	1.747	1.789	4.475
Euribor: SD	1.749	0.718	1.743
Euribor: autocorrelation	0.979	0.848	0.972
Euribor: share ≤ 0	0.226	0.000	0.107

Notes: Quantiles are based on 1,000 unconditional stationary replications of 84 quarters. Autocorrelation is at lag one; rate shares are fractions. Observable units follow Section 2.

Observed GDP-growth mean, volatility and autocorrelation fall within the central replication intervals. However, none of the replications produces a minimum as low as the observed −3.094%. The extreme contraction is therefore unusual under the fitted model, although it is not impossible.

Replicated inflation is more volatile and persistent than observed inflation: both observed statistics lie below their fifth replication percentiles. For Euribor, the observed mean is near the lower tail, while volatility and persistence are near or beyond the upper ends of the central intervals. The observed share of non-positive rates is 22.6%, compared with a 95th replication percentile of 10.7%. The model consequently struggles to reproduce the low-rate period alongside the other dynamics.

In-sample filtering diagnostics also indicate systematic interest-rate overprediction at the MAP estimate. Their definitions and results are documented in Appendix C.3.

4.5 Sensitivity to fixed calibration

Sensitivity is examined by changing β to 0.995 or 0.9999, or σ to 1.5 or 2, one at a time. All eleven estimated parameters are reoptimised under each calibration,

keeping the univariate priors and bounds unchanged. The complete results are reported in Table 14 in Appendix C.2.

Increasing β to 0.9999 reduces the fitted long-run rate from 3.042 to 2.791%, but leaves a gap of approximately 1.04 percentage points relative to the observed mean. Rate-error autocorrelation remains largely unchanged. The lower β and higher σ variants produce larger rate gaps, so the examined changes do not resolve the mismatch.

These are sensitivity exercises at reoptimised MAP estimates, without additional posterior sampling or marginal-likelihood comparisons. The baseline calibration is retained for forecasting; test-period observations are not used to select a variant.

5 The BVAR and Forecasting Experiment

5.1 BVAR specification and prior

The comparison model is a three-variable Bayesian vector autoregression with two quarterly lags, denoted BVAR(2). It uses the same observables and estimation sample as the DSGE model. Bayesian shrinkage regularises the lag coefficients, following the approach reviewed by Karlsson [2013]. The lag order and prior settings are specified before evaluating test-period forecasts.

Each observable is standardised using its estimation-sample mean and sample standard deviation:

$$\widetilde{z}_{j,t} = \frac{z_{j,t} - \overline{z}_{j,\text{est}}}{d_{j,\text{est}}}. \tag{31}$$

These scaling constants remain fixed during evaluation, and forecasts are converted back to the original units. The model is

$$\widetilde{z}_t = c + A_1 \widetilde{z}_{t-1} + A_2 \widetilde{z}_{t-2} + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \Sigma). \tag{32}$$

Coefficients and innovation covariance are constant. After allocating two observations to initial lags, 82 regression observations remain.

Let B collect the intercept and lag coefficients in a 7×3 matrix. A conjugate normal–inverse-Wishart prior is used:

$$\mathsf{B} \mid \Sigma \sim \mathcal{MN}_{7\times 3}(\mathsf{B}_0, V_0, \Sigma), \tag{33}$$

$$\Sigma \sim \mathcal{IW}(S_0, 5). \tag{34}$$

The own-first-lag prior means are $(0, 0.8, 0.95)$ for growth, inflation and the interest rate, respectively. All other coefficient prior means are zero. The row covariance is

$$V_0 = \operatorname{diag}\left(100, \tau^2, \tau^2, \tau^2, \tau^2/4, \tau^2/4, \tau^2/4\right), \qquad \tau = 0.2. \tag{35}$$

Thus, second-lag coefficients receive tighter shrinkage. The scale matrix S_0 is diagonal, with entries equal to residual variance estimates from separate AR(1)

regressions with intercepts fitted to the standardised estimation data. With three variables and five inverse-Wishart degrees of freedom, $\mathbb{E}[\Sigma] = S_0$.

The prior is Minnesota-inspired rather than an exact classical Minnesota specification: conjugacy couples coefficient uncertainty across equations through Σ, tightness is fixed, and the covariance scale is calibrated from training data. The analytical posterior supplies 4,000 independent joint draws without MCMC. Regression matrices, posterior updates and sampling formulas are documented in Appendix B.5.

Posterior draws are not truncated to impose stationarity. Their companion-matrix roots are retained as a diagnostic. Forecasts are conditional on observations at each origin and require no structural identification of BVAR innovations.

5.2 Benchmark forecasts

The growth benchmark is the fixed 1999–2019 sample mean, while inflation and interest-rate benchmarks use the latest observation at each origin o:

$$\widehat{g}^{\text{bench}}_{o+h|o} = \frac{1}{84} \sum_{t \in \mathcal{T}_{\text{est}}} g_t^{\text{obs}}, \tag{36}$$

$$\widehat{\pi}^{\text{bench}}_{o+h|o} = \pi_o^{\text{obs}}, \tag{37}$$

$$\widehat{i}^{\text{bench}}_{o+h|o} = i_o^{\text{obs}}, \qquad h \in \{1, 4\}. \tag{38}$$

The growth mean is not updated during evaluation. These benchmarks are evaluated only for point accuracy; no predictive distributions are assigned to them.

5.3 Forecast origins, horizons and updating rules

Forecasts target the individual quarterly observation $o + h$ at horizons $h = 1$ and $h = 4$. In particular, the four-quarter growth forecast concerns quarter-on-quarter growth in the fourth future quarter, not cumulative annual growth.

Both models are estimated through 2019Q4. Their parameter distributions and posterior-draw weights remain fixed thereafter. At each origin, observations through that date update the DSGE filtered state distribution separately for each parameter draw and supply the two latest BVAR lag vectors. No future observations are used for state smoothing, and test data do not select model settings.

Later observations do not reweight DSGE parameter draws by their subsequent likelihoods or update the BVAR parameter posterior. The predictive distributions therefore combine origin-specific states or lags with the pre-2020 posterior, rather than performing a full Bayesian parameter update at every origin.

Table 5 shows the resulting evaluation samples. Origins before 2019Q4 are excluded because the parameter estimates would otherwise use observations after the origin. The two horizons have different target samples, which matters when comparing their performance.

Table 5: Forecast origins and target samples

Horizon	Origins	Targets	N_h
$h = 1$	2019Q4–2025Q3	2020Q1–2025Q4	24
$h = 4$	2019Q4–2024Q4	2020Q4–2025Q4	21

Results are also separated by target quarter into 2020–2021 and 2022–2025. The earlier subperiod contains eight one-quarter targets and five four-quarter targets; the later period contains sixteen at each horizon. The exercise uses revised data and assumes observations through the origin are available, without reproducing historical publication delays.

The DSGE forecasts use 2,000 posterior draws and the BVAR forecasts use 4,000. Draw selection and random seeds are documented in Appendix D.

5.4 Predictive distributions and evaluation metrics

DSGE predictive simulations integrate over posterior parameter draws, the filtered state distribution and future structural innovations. BVAR simulations integrate over coefficient and innovation-covariance draws and future innovations, starting from observed lags. Parameters remain constant within each simulated path.

For both models, point forecasts average the draw-specific conditional means, calculated without simulated future innovations. The central 90% predictive interval uses the fifth and ninety-fifth percentiles of simulated observations. These intervals describe future observations, rather than model parameters. They are conditional on the chosen specifications and, for the DSGE, the fixed calibration.

Table 6 summarises the evaluation measures. Interval score and CRPS assess probabilistic forecasts using the scoring-rule framework of Gneiting and Raftery [2007]. Simulation recursions and exact metric definitions are provided in Appendix C.4.

Scores are compared across models for the same variable, horizon and target sample. CRPS evaluates each observable separately, rather than the joint predictive distribution. Coverage alone is insufficient, since wide intervals can cover observations while providing limited precision.

The short evaluation sample and overlapping four-quarter forecast windows limit inference about model rankings. Results are reported descriptively, without formal forecast-ranking significance tests or test-period retuning.

6 Results

The comparisons below are descriptive rankings for the specified models and evaluation samples. Complete point and probabilistic forecast metrics are reported in Appendix C.1.

Table 6: Forecast evaluation measures

Measure	Interpretation
RMSE	Root mean squared error; gives greater weight to large errors. Lower is better.
MAE	Mean absolute error. Lower is better.
Bias	Mean forecast minus observation. Negative values indicate underprediction.
Coverage	Fraction of observations within the central 90% predictive interval.
Interval width	Average distance between interval bounds; interpreted jointly with coverage.
Interval score	Penalises interval width and observations outside the interval. Lower is better.
CRPS	Evaluates the full marginal predictive distribution. Lower is better.

6.1 Point forecast performance

Table 7 reports full-sample RMSE. Comparisons are made within each observable and horizon. The horizons cover different target samples, as specified in Section 5.3; their RMSE values therefore do not isolate the effect of forecast horizon.

Table 7: Point forecast performance: full evaluation sample

Observable	Horizon	DSGE	BVAR	Benchmark
GDP growth	$h = 1$	4.346	4.055	**3.415**
Inflation	$h = 1$	**2.158**	2.674	2.191
Euribor	$h = 1$	0.551	0.948	**0.443**
GDP growth	$h = 4$	**0.358**	0.551	0.577
Inflation	$h = 4$	**3.341**	3.914	4.362
Euribor	$h = 4$	**1.426**	2.439	1.648

Notes: RMSE is in the original observable units. Bold identifies the lowest value in each row, without implying statistical significance. There are 24 targets at $h = 1$ and 21 at $h = 4$. Benchmark rules are defined in Section 5.2.

The DSGE model has lower RMSE than the BVAR in five of six comparisons and the lowest RMSE among all three approaches for every observable at $h = 4$. At $h = 1$, however, the historical-mean benchmark performs best for growth and the last-observation benchmark performs best for Euribor.

The DSGE's one-quarter inflation advantage over the benchmark is small and depends on the loss function: RMSE is 2.158 versus 2.191, while MAE favours the benchmark, at 1.752 versus 1.807. Both estimated models underpredict inflation on average. DSGE and BVAR biases are -0.496 and -1.148 at $h = 1$, and -1.629 and -2.436 at $h = 4$.

6.2 Predictive interval performance

Table 8 evaluates central 90% predictive intervals and marginal predictive distributions.

Table 8: Predictive interval and distributional performance

Observable	Model	Coverage (%)	Width	Interval score	CRPS
One-quarter horizon: $h = 1$					
GDP growth	DSGE	79.2	1.880	30.030	1.699
	BVAR	79.2	2.024	25.610	1.531
Inflation	DSGE	70.8	4.377	10.626	1.301
	BVAR	75.0	4.745	15.320	1.517
Euribor	DSGE	70.8	1.182	2.662	0.314
	BVAR	54.2	1.101	5.157	0.488
Four-quarter horizon: $h = 4$					
GDP growth	DSGE	100.0	2.327	2.327	0.233
	BVAR	95.2	2.026	2.692	0.263
Inflation	DSGE	71.4	6.098	23.172	1.833
	BVAR	71.4	4.825	30.393	2.228
Euribor	DSGE	81.0	2.807	7.125	0.772
	BVAR	61.9	3.304	14.167	1.349

Notes: Nominal coverage is 90%. Width, interval score and CRPS are averages in the observable's original units. Lower scores are better. Benchmarks have no assigned predictive distributions.

Inflation and interest-rate coverage falls below 90% for both models at both horizons. The DSGE achieves higher rate coverage and lower probabilistic losses than the BVAR, but still misses a substantial share of outcomes.

Coverage rankings need not agree with scoring-rule rankings. For one-quarter inflation, the BVAR has higher coverage but wider intervals and worse interval score and CRPS. At $h = 4$, inflation coverage is identical, while the DSGE's wider intervals accompany lower scores. Four-quarter GDP coverage is high, particularly for the DSGE, whose interval score equals its width because every outcome is covered. This does not establish perfect predictive calibration.

Overall, the DSGE has lower interval score and CRPS in five of six comparisons; one-quarter growth is the exception. Figure 2 shows the missed inflation surge and large model rate forecasts following the extreme output observations in 2020.

6.3 Performance across subperiods

Table 9 separates the evaluation by target quarter. The 2020–2021 subsample contains eight targets at $h = 1$ and five at $h = 4$; the later subsample contains sixteen at each horizon.

Pandemic observations dominate the one-quarter growth errors. The targets 2020Q2–2020Q4 account for 96.6% of the DSGE's full-sample squared growth

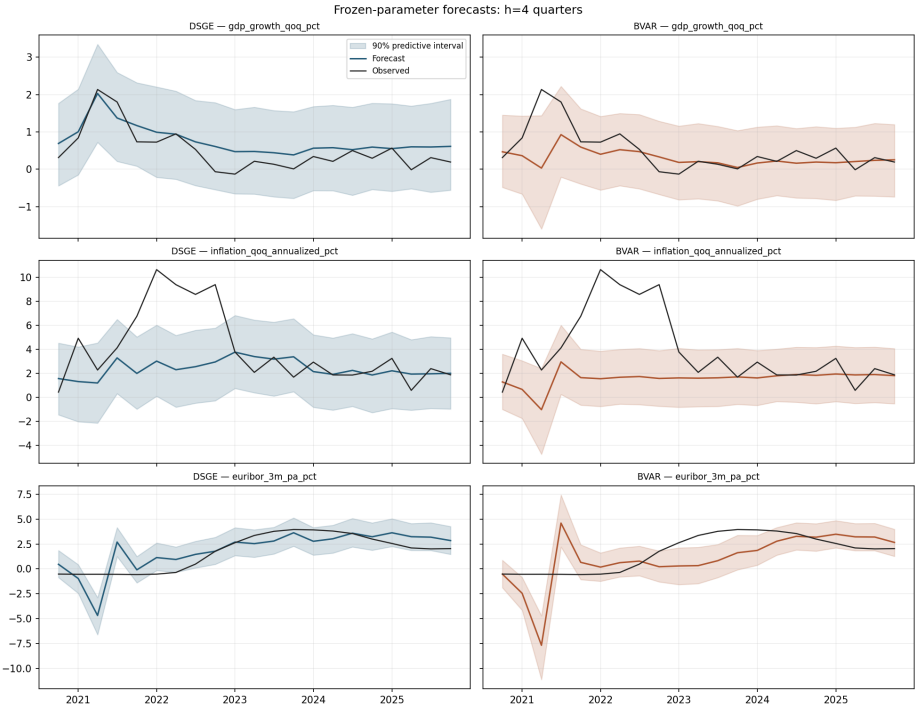


Figure 2: Four-quarter forecasts and central 90% predictive intervals. The horizontal axis gives target quarters. Parameter distributions and posterior weights remain fixed after 2019Q4; states and observed lags are updated at each origin.

Table 9: RMSE by evaluation subperiod

		2020–2021			2022–2025		
Observable	h	DSGE	BVAR	Benchmark	DSGE	BVAR	Benchmark
GDP growth	1	7.517	7.013	**5.900**	0.281	**0.278**	0.293
Inflation	1	2.375	2.607	**2.335**	**2.040**	2.707	2.115
Euribor	1	0.647	1.504	**0.074**	0.496	**0.465**	0.541
GDP growth	4	**0.333**	1.045	1.060	0.366	**0.239**	0.293
Inflation	4	**2.793**	3.391	4.381	**3.494**	4.064	4.356
Euribor	4	2.411	4.067	**0.140**	**0.923**	1.624	1.886

Notes: Bold identifies the lowest RMSE within each subperiod and row. Four-quarter targets begin in 2020Q4.

forecast error and 95.8% of the BVAR's. For example, the DSGE forecasts -5.860% growth for 2020Q3, against an observed rebound of 10.919%. The fixed-mean benchmark avoids extrapolating the contraction and has the lowest growth RMSE during 2020–2021, although it does not anticipate the pandemic movements.

During 2022–2025, one-quarter growth accuracy is similar for the estimated models. At $h = 4$, the BVAR outperforms both the DSGE and benchmark, reversing the full-sample ranking. The DSGE's later four-quarter growth bias is 0.306, compared with -0.051 for the BVAR, indicating a poorer alignment of its forecast level with observed growth.

The last-observation rate benchmark performs particularly well during 2020–2021, when rates change little. During 2022–2025, the BVAR has the lowest one-quarter rate RMSE, while the DSGE performs best at four quarters and for inflation at both horizons.

Growth interval performance also changes sharply: one-quarter coverage is 37.5% for both estimated models in 2020–2021 and 100% in 2022–2025. Full-sample coverage therefore combines missed extremes with complete coverage during a less volatile period.

6.4 Structural impulse responses

Figure 3 reports posterior median DSGE responses over 40 quarters, with pointwise 90% credible intervals from 2,000 parameter draws. Subsequent innovations are set to zero, so the bands reflect parameter uncertainty conditional on the model and fixed calibration. They are neither simultaneous bands nor predictive intervals.

Preference and cost-push disturbances are positive one-standard-deviation shocks for each parameter draw. The monetary disturbance is normalised to raise the annualised rate by 25 basis points at impact. This fixes the total contemporaneous rate response, including endogenous feedback, rather than the additive policy-rule innovation. The output response is a level deviation from deterministic trend, not quarterly GDP growth. Calculation details are given in Appendix A.6.

The monetary disturbance lowers output and inflation, with the median output contraction reaching approximately 0.083% two quarters after impact. The impact rate response has no posterior dispersion because its magnitude is imposed by normalisation. The cost-push disturbance raises inflation and the policy rate while reducing output, with persistent effects reflecting both shock persistence and endogenous propagation.

The preference disturbance increases output, but its median inflation response is initially negative. Its impact inflation interval includes zero. This pattern follows from the shock's joint influence on the Euler condition and marginal cost, including the direct term $-b_t$ in Equation (9). It should therefore be interpreted as the model's specific preference disturbance rather than a generic positive demand shock.

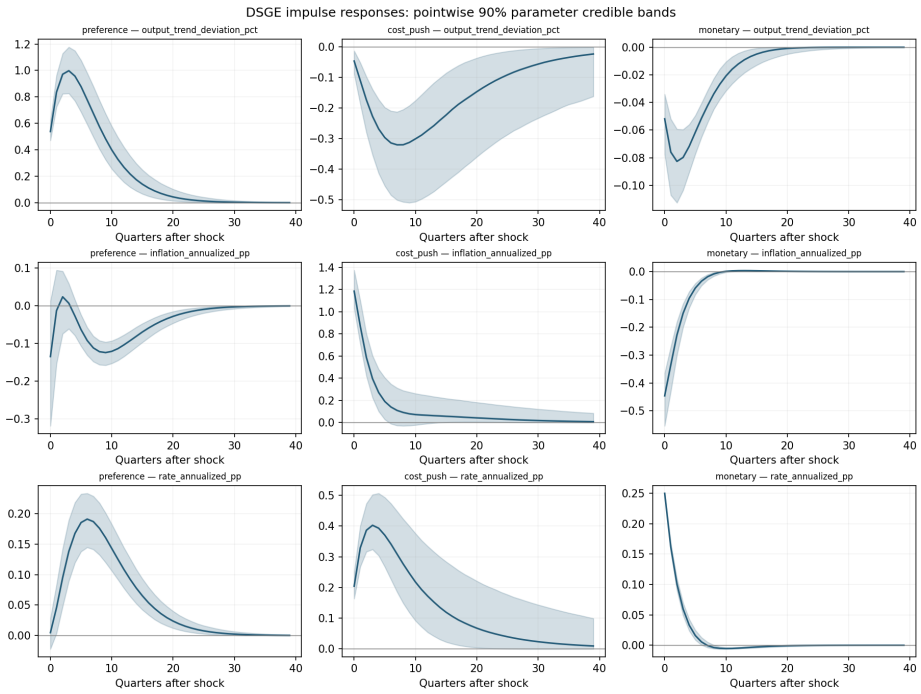


Figure 3: DSGE impulse responses. Lines show posterior medians and shading the pointwise 5th–95th percentiles. Preference and cost-push shocks are one-standard-deviation innovations; the monetary shock produces a 25-basis-point impact rate increase. Output is the log-percent level deviation from trend; inflation and rate responses are in annualised percentage points.

These responses describe transmission within the maintained model; they are not externally identified causal estimates. Their interpretation is considered alongside the validation evidence in Section 7.

7 Discussion and Limitations

7.1 Structural interpretation and forecast performance

Forecast accuracy, empirical fit and structural identification address different questions. The forecast comparison assesses predictive performance under the information set and updating rules specified in Section 5. The posterior predictive checks instead assess whether the DSGE model reproduces selected properties of the estimation data. Favourable forecast results can coexist with the discrepancies documented in Section 4.4.

The DSGE structure connects observable dynamics to explicit economic mechanisms and permits the conditional impulse-response analysis in Section 6.4. However, these responses depend on the maintained equations, shock processes and calibration. Numerical verification and satisfactory MCMC diagnostics support the reliability of the calculations; they do not establish that the shocks are uniquely identified disturbances in the actual economy. The responses should therefore be interpreted as model-implied effects, rather than externally identified causal estimates.

Economic restrictions may help forecasting by regularising the dynamics, but this experiment does not establish why a particular model performs better. Its results concern the two fitted specifications and the chosen benchmarks. The variation across variables, horizons and subperiods does not support a general DSGE–BVAR ranking.

7.2 Limitations of the experiment

Evaluation design. The short test period contains unusually large macroeconomic disturbances, and the subperiod comparisons rely on few observations. Consequently, individual quarters can substantially affect the reported losses. Four-quarter-ahead forecasts from consecutive origins also have overlapping forecast windows, so their errors may be serially dependent. Moreover, the horizons cover different target samples, as shown in Section 5.3. Differences across horizons therefore reflect both forecast distance and sample composition. The comparisons are descriptive; statistical significance of the rankings has not been assessed.

Data and measurement. The experiment uses revised data and assumes that complete quarterly observations are available at each origin. It does not reproduce historical data vintages, publication delays or incomplete information sets, and should therefore be interpreted as a pseudo-out-of-sample exercise

rather than a real-time evaluation. The geographic mismatch documented in Section 2 introduces a further comparability limitation.

Euribor also contains market-rate components that the DSGE model's single nominal interest rate does not separately represent. Without an explicit spread, the model may attribute movements in those components to its existing states and shocks. In addition, the steady-state Euler restriction links the rate level to trend growth, trend inflation and fixed preference parameters. The sensitivity analysis shows that changing this calibration affects the rate level without resolving all discrepancies in its dynamics.

Model specification and parameter stability. The small, first-order DSGE model described in Section 3 omits mechanisms that may matter during large disruptions. Its linear dynamics and constant-variance Gaussian shocks limit its representation of extreme observations, changing volatility and nonlinear policy constraints.

Both models retain their pre-2020 parameter distributions throughout the evaluation. New observations update DSGE states and BVAR lags, but not parameter values or posterior weights. This isolates the performance of the original estimates while preventing adaptation through parameter learning. DSGE inference is also conditional on the fixed calibration. The re-estimated MAP sensitivity exercises are neither full posterior comparisons nor a treatment of calibration uncertainty.

Priors and uncertainty. The BVAR lag length and shrinkage settings are predetermined rather than systematically compared. Its results therefore apply to this particular specification. The posterior is not restricted to stable coefficient draws: approximately 5.6% of sampled draws have a companion-matrix spectral radius of at least one. All reported forecast simulations are finite, but these draws can affect predictive tails. Imposing stability would change the posterior specification.

The predictive distributions include the uncertainty components described in Section 5.4, but exclude uncertainty about model choice, omitted mechanisms, data revisions and structural breaks. Fixed calibration introduces an additional omission for the DSGE forecasts. Their intervals consequently measure uncertainty within the fitted models rather than all sources of forecast uncertainty.

7.3 Possible extensions

The following extensions have not been implemented.

First, a real-time evaluation could use historical data vintages and publication calendars at each forecast origin. This would distinguish the effects of revisions and delayed releases from those of the forecasting models, while requiring explicit treatment of incomplete quarterly information.

Second, recursive estimation could assess whether updating parameter distributions improves forecasts relative to updating states and lags alone. An

expanding-window design would provide a direct comparison with the present protocol. Updating rules should be specified using training-period validation or evaluated on a new test sample, rather than selected to improve the results already observed here.

Third, time-varying innovation variances could address the limitations of constant-volatility predictive distributions. A BVAR with stochastic volatility would offer a focused extension, although it would require a different estimation procedure from the conjugate approach used here. Changing volatility would not, by itself, resolve misspecified economic dynamics.

A further targeted extension would introduce an explicit spread between Euribor and the model's policy-related rate, supported by an additional interest-rate series. This could help separate market-rate components from monetary policy dynamics, subject to renewed assessment of identification and estimation diagnostics.

8 Conclusion

This study compared a small estimated New Keynesian DSGE model with a BVAR and simple benchmarks for forecasting euro-area output growth, inflation and the short-term interest rate. Relative performance varies across variables, horizons and subperiods. Over the full evaluation period, the DSGE model has lower RMSE than the BVAR in five of the six variable–horizon combinations and the lowest RMSE among the three approaches for one-quarter-ahead inflation and all three four-quarter-ahead forecasts. Nevertheless, the benchmarks perform best for one-quarter-ahead growth and interest rates. Neither estimated model therefore consistently dominates the alternatives.

These comparisons depend on the episodes included in the evaluation. Pandemic quarters dominate the one-quarter-ahead growth errors, while the BVAR achieves the lowest growth RMSE at both horizons during 2022–2025. Predictive interval evaluation and model validation also qualify the point forecast results. Both models frequently undercover inflation and interest rates, and the DSGE posterior predictive checks reveal discrepancies in extreme growth outcomes, inflation volatility and persistence, and the interest-rate distribution. Calibration sensitivity further highlights the importance of the steady-state rate restriction. Satisfactory numerical checks and convergence diagnostics thus coexist with substantive limitations in empirical fit.

The project's contribution is a coherent applied Python workflow combining model solution, Bayesian estimation, validation and forecast evaluation. The DSGE model additionally provides an economic interpretation of conditional shock propagation, while the BVAR supplies a complementary forecasting comparison. That structural interpretation remains conditional on the maintained equations, shock definitions and fixed calibration; favourable forecasts do not establish causal identification. Taken together, the findings illustrate the value of assessing forecast accuracy, predictive calibration and empirical adequacy jointly. They support conclusions about these fitted specifications under the

chosen protocol, rather than a general ranking of DSGE and BVAR methods.

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A Model Derivation and Solution

A.1 Economic environment and balanced growth

Under the environment described in Section 3.1, aggregate consumption equals output, $C_t = Y_t$, and productivity follows $A_t = GA_{t-1}$. Household preferences are

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[e^{b_t} \frac{(C_t - h_0 C_{t-1}^a)^{1-\sigma}}{1-\sigma} - \chi A_t^{1-\sigma} \frac{N_t^{1+\varphi}}{1+\varphi} \right], \tag{39}$$

with the logarithmic limit used when $\sigma = 1$. Lagged aggregate consumption C_{t-1}^a is external to the household's optimisation problem; in equilibrium, $C_{t-1}^a = C_{t-1}$. Habit formation and price indexation are standard mechanisms in New Keynesian modelling [Christiano et al., 2005], although the present specification is not a replication of that model.

Let Q_t denote nominal bond holdings paying one unit per unit held in period $t+1$, and let $\mathcal{D}_t$ denote nominal profits. The household budget constraint is

$$P_t C_t + \frac{Q_t}{R_t} = W_t N_t + Q_{t-1} + \mathcal{D}_t.$$

With real wage $w_t = W_t / P_t$, the first-order conditions are

$$\Lambda_t = e^{b_t} (C_t - h_0 C_{t-1})^{-\sigma}, \tag{40}$$

$$\Lambda_t = \beta R_t \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Pi_{t+1}} \right], \tag{41}$$

$$w_t \Lambda_t = \chi A_t^{1-\sigma} N_t^{\varphi}. \tag{42}$$

External habit does not introduce the future-utility derivative associated with internal habit.

Intermediate producers use $Y_{j,t} = A_t N_{j,t}$. The CES final-good aggregator and associated demand are

$$Y_t = \left[\int_0^1 Y_{j,t}^{(\epsilon-1)/\epsilon} \, dj \right]^{\epsilon/(\epsilon-1)}, \qquad Y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\epsilon} Y_t,$$

where $\epsilon > 1$. Real marginal cost is $MC_t = w_t/A_t$. The steady-state markup is $\epsilon/(\epsilon-1)$; ϵ is not separately estimated in the implemented first-order system.

The productivity factor in labour disutility permits a balanced-growth steady state with constant labour and consumption growing at rate G. The steady-state Euler condition gives

$$\bar{R} = \frac{\bar{\Pi} G^{\sigma}}{\beta}. \tag{43}$$

A.2 Detrending and household equations

Use the deviations y_t, π_t and r_t defined in Section 3.2. Writing $\widetilde{C}_t = C_t/A_t$ gives

$$C_t - h_0 C_{t-1} = A_t \left(\widetilde{C}_t - h \widetilde{C}_{t-1} \right), \qquad h = \frac{h_0}{G}, \qquad a = \frac{\sigma}{1-h}.$$

Consequently, the first-order deviation of detrended marginal utility is

$$\lambda_t = b_t - a(y_t - h y_{t-1}). \tag{44}$$

The log-linear Euler condition is

$$\lambda_t = \mathbb{E}_t \lambda_{t+1} + r_t - \mathbb{E}_t \pi_{t+1}.$$

Substitution, together with $\mathbb{E}_t b_{t+1} = \rho_b b_t$, yields

$$a(1+h) y_t = a \mathbb{E}_t y_{t+1} + a h y_{t-1} - r_t + \mathbb{E}_t \pi_{t+1} + (1-\rho_b) b_t. \tag{45}$$

Price dispersion has no first-order effect on aggregate labour around the steady state. Production and labour supply therefore imply

$$mc_t = \varphi y_t - \lambda_t,$$

and hence

$$mc_t = (\varphi + a) y_t - a h y_{t-1} - b_t. \tag{46}$$

Here mc_t is the log deviation of real marginal cost. These equations concern output relative to its deterministic trend, rather than output relative to a flexible-price allocation.

A.3 Price setting and monetary policy

Under Calvo pricing, a producer resets its price with probability $1-\theta$ [Calvo, 1983]. Otherwise,

$$P_{j,t} = P_{j,t-1} \bar{\Pi} \left(\frac{\Pi_{t-1}}{\bar{\Pi}} \right)^{\iota}.$$

For a price selected at time t, cumulative indexation is

$$I_{t,t+k} = \prod_{\ell=1}^{k} \bar{\Pi} \left(\frac{\Pi_{t+\ell-1}}{\bar{\Pi}} \right)^{\iota}, \qquad I_{t,t} = 1.$$

The reset-price condition is

$$\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \frac{\Lambda_{t+k}}{\Lambda_t} Y_{j,t+k} \left[\epsilon MC_{t+k} - (\epsilon - 1) \frac{P_t^* I_{t,t+k}}{P_{t+k}} \right] = 0. \tag{47}$$

The aggregate price index satisfies

$$P_t^{1-\epsilon} = (1-\theta)(P_t^*)^{1-\epsilon} + \theta \left[P_{t-1} \bar{\Pi} \left(\frac{\Pi_{t-1}}{\bar{\Pi}} \right)^{\iota} \right]^{1-\epsilon}.$$

The balanced-growth discount factor relevant for price setting is

$$B = \beta G^{1-\sigma}, \qquad 0 < B < 1,$$

because marginal utility falls at rate $G^{-\sigma}$ while demand grows at rate G. Let x_t denote the log deviation of the reset price relative to the aggregate price level. Linearisation gives

$$(1-\theta)x_t = \theta(\pi_t - \iota\pi_{t-1}), \tag{48}$$

$$x_t = (1 - B\theta)mc_t + B\theta\mathbb{E}_t \left[x_{t+1} + \pi_{t+1} - \iota\pi_t \right]. \tag{49}$$

Eliminating x_t yields the indexed Phillips curve. Adding the implemented cost-push disturbance gives

$$(1 + B\iota)\pi_t = B\mathbb{E}_t\pi_{t+1} + \iota\pi_{t-1} + \kappa mc_t + (1 + B\iota)u_t, \tag{50}$$

where

$$\kappa = \frac{(1-\theta)(1-B\theta)}{\theta}.$$

Thus u_t enters the normalised inflation equation with coefficient one.

The interest-rate feedback rule and the three shock processes are specified in Section 3.2. Their coefficients enter the complete system below. The policy rule follows the interest-rate feedback approach associated with Taylor [1993], with detrended output as its output argument.

A.4 Complete linear system and matrix representation

Using the state and decision ordering from Section 3.3,

$$s_t = (y_{t-1}, \pi_{t-1}, r_{t-1}, b_t, u_t, m_t)', \qquad d_t = (y_t, \pi_t, r_t)',$$

define $v_t = (s_t', d_t')'$ and $D = 1 + B\iota$. The complete conditional expectation system is

$$M\mathbb{E}_t v_{t+1} = N v_t,$$

with

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

and

$$N = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \rho_b & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_u & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -ah & 0 & 0 & -(1-\rho_b) & 0 & 0 & a(1+h) & 0 & 1 \\ \kappa ah & -\iota & 0 & \kappa & -D & 0 & -\kappa(\varphi+a) & D & 0 \\ 0 & 0 & -\rho_r & 0 & 0 & -1 & -(1-\rho_r)\phi_y & -(1-\rho_r)\phi_\pi & 1 \end{pmatrix}.$$

Rows one to three update endogenous lags; rows four to six describe expected shock propagation. The final rows impose the Euler equation, Phillips curve and policy rule.

An ordered generalised Schur decomposition of (N, M) provides the stable invariant subspace [Klein, 2000]. Exactly six roots must be stable. Partition its Schur vectors as

$$Z_{\text{stable}} = \begin{pmatrix} Z_s \\ Z_d \end{pmatrix}, \qquad Z_s \in \mathbb{C}^{6\times 6}, \quad Z_d \in \mathbb{C}^{3\times 6}.$$

When Z_s is nonsingular, the decision rule is

$$d_t = H s_t, \qquad H = Z_d Z_s^{-1}.$$

The accepted solution must be real within numerical tolerance. The static policy equation generates an infinite generalised eigenvalue.

Partition $H = (H_\ell \; H_{\text{sh}})$ into two 3×3 blocks and define

$$R_{\text{sh}} = \text{diag}(\rho_b, \rho_u, 0).$$

The state transition matrices are

$$F = \begin{pmatrix} H_\ell & H_{\text{sh}} \\ 0_{3\times 3} & R_{\text{sh}} \end{pmatrix}, \qquad J = \begin{pmatrix} 0_{3\times 3} \\ \text{diag}(s_b, s_u, s_m) \end{pmatrix}. \tag{51}$$

Thus $s_{t+1} = F s_t + J\varepsilon_{t+1}$.

With $T = (I_6 \; H')'$, verification requires

$$MTF - NT = 0, \qquad \rho(F) < 1.$$

The implementation rejects roots within 10^{-8} of the unit circle, a state-subspace condition number above 10^{10}, or a solution residual above 10^{-8}.

A.5 Observation system

For the observed vector defined in Section 3.3, write

$$z_t = \mu + C s_t, \qquad C = \begin{pmatrix} 100(H_{1,\cdot} - e_1') \\ 400 H_{2,\cdot} \\ 400 H_{3,\cdot} \end{pmatrix}, \qquad \mu = \begin{pmatrix} \bar{g} \\ \bar{\pi} \\ \bar{i} \end{pmatrix},$$

where e_1 selects lagged output from the state vector. The steady-state condition (43) implies

$$\bar{g} = 100 \log G, \qquad \bar{\pi} = 400 \log \bar{\Pi}, \qquad \bar{i} = \bar{\pi} + 400(\sigma \log G - \log \beta).$$

There is no independently estimated rate intercept. The rate measurement uses the annualised log-rate approximation to Euribor described in the main text. No measurement error is added. Stationary covariance initialisation is documented in Appendix B.2.

A.6 Impulse-response computation

For parameter draw $\vartheta^{(r)}$, a shock of magnitude $a_k^{(r)}$ to innovation k generates

$$\Delta s_\ell^{(r,k)} = \left(F^{(r)}\right)^\ell J^{(r)} e_k a_k^{(r)}, \qquad \ell = 0, \ldots, 39.$$

Responses of output relative to trend, annualised inflation and the annualised rate are

$$\mathrm{IRF}_\ell^{(r,k)} = \mathrm{diag}(100, 400, 400) H^{(r)} \left(F^{(r)}\right)^\ell J^{(r)} e_k a_k^{(r)}. \tag{52}$$

The output response is a log-level response relative to trend, not an observed growth response.

Preference and cost-push shocks use $a_k^{(r)} = 1$, corresponding to one innovation standard deviation under each draw. For the monetary shock,

$$a_m^{(r)} = \frac{0.25}{400 H_{3,\cdot}^{(r)} J^{(r)} e_m}.$$

This fixes the total impact rate response at 0.25 percentage points in every draw, including contemporaneous policy feedback.

Medians and pointwise 5th and 95th percentiles are calculated across the 2,000 selected posterior draws. The bands represent parameter uncertainty conditional on the calibration, with no additional future innovations.

B Priors and Computational Diagnostics

B.1 Complete parameter and prior specification

The fixed calibration is given in Section 4.1. Table 10 reports the eleven estimated parameters. Prior settings are project choices; the Bayesian estimation framework follows An and Schorfheide [2007].

Table 10: Estimated parameters, parent priors and bounds

Parameter	Interpretation	Parent prior	Bounds
h_0	External habit	Beta: mean 0.6, SD 0.2	$(0.001, 0.95)$
ι	Inflation indexation	Beta: mean 0.3, SD 0.2	$(0.001, 0.95)$
ρ_r	Rate smoothing	Beta: mean 0.8, SD 0.1	$(0.001, 0.98)$
ϕ_π	Inflation feedback	$1 + \text{Gamma}(4, 0.125)$	$(1.001, 4)$
ρ_b	Preference persistence	Beta: mean 0.7, SD 0.15	$(0.001, 0.98)$
ρ_u	Cost-push persistence	Beta: mean 0.5, SD 0.2	$(0.001, 0.98)$
s_b	Preference shock SD	$\log s_b \sim N(\log 0.005, 0.8^2)$	$(10^{-5}, 0.2)$
s_u	Cost-push shock SD	$\log s_u \sim N(\log 0.001, 0.8^2)$	$(10^{-5}, 0.05)$
s_m	Monetary shock SD	$\log s_m \sim N(\log 0.001, 0.8^2)$	$(10^{-5}, 0.05)$
$\bar{g}$	Quarterly trend growth	$N(0.3, 0.2^2)$	$(-0.5, 1)$
$\bar{\pi}$	Annualised trend inflation	$N(2, 0.75^2)$	$(0, 5)$

For a Beta parent distribution with mean m and SD s, the shape parameters are

$$c = \frac{m(1-m)}{s^2} - 1, \qquad \alpha = mc, \qquad \delta = (1-m)c.$$

The Gamma prior uses shape and scale parametrisation. The means and SDs in the table refer to parent distributions before truncation.

For parent density f_j and CDF F_j, the bounded density is

$$p_j(\vartheta_j) = \frac{f_j(\vartheta_j)}{F_j(U_j) - F_j(L_j)} \mathbf{1}\{L_j < \vartheta_j < U_j\}.$$

The joint prior additionally restricts parameters to the admissible solution region $\mathcal{A}$:

$$p(\vartheta) \propto \prod_j p_j(\vartheta_j) \mathbf{1}\{\vartheta \in \mathcal{A}\}.$$

Its admissibility normalising constant is unnecessary for the within-specification posterior calculations performed here.

B.2 Likelihood and posterior computation

The Gaussian likelihood uses the Kalman filter [Kalman, 1960]. For each admissible parameter vector, stationary initialisation is

$$a_{1|0} = 0, \qquad P_{1|0} = P_\infty, \qquad P_\infty = F P_\infty F' + JJ'.$$

The discrete Lyapunov equation is solved separately for each parameter vector.

Given predicted state mean $a_{t|t-1}$ and covariance $P_{t|t-1}$, the innovation and its covariance are

$$v_t = z_t - \mu - C a_{t|t-1}, \qquad S_t = C P_{t|t-1} C'.$$

The update and next-period prediction are

$$K_t = P_{t|t-1} C' S_t^{-1}, \tag{53}$$

$$a_{t|t} = a_{t|t-1} + K_t v_t, \tag{54}$$

$$P_{t|t} = (I - K_t C) P_{t|t-1} (I - K_t C)', \tag{55}$$

$$a_{t+1|t} = F a_{t|t}, \tag{56}$$

$$P_{t+1|t} = F P_{t|t} F' + JJ'. \tag{57}$$

The covariance update is the Joseph form with zero measurement error covariance. Cholesky factorisation of S_t is used for linear solves and log determinants, without added covariance jitter.

With three observables and T_{est} observations,

$$\log L(\vartheta) = -\frac{1}{2} \sum_{t=1}^{T_{\text{est}}} \left[3 \log(2\pi) + \log |S_t| + v_t' S_t^{-1} v_t \right].$$

Invalid model solutions or failed covariance factorisations receive zero posterior density.

For bounds (L_j, U_j), define

$$q_j = \text{logit}^{-1}(\eta_j), \qquad \vartheta_j = L_j + (U_j - L_j) q_j.$$

The transformed log posterior is

$$\ell_\eta(\eta) = \log L(\vartheta(\eta)) + \log p(\vartheta(\eta)) + \sum_j \left[\log(U_j - L_j) + \log q_j + \log(1 - q_j) \right].$$

MAP optimisation instead maximises the posterior density in physical parameter coordinates, without the transformation Jacobian. Two starting points are used with L-BFGS-B.

Random-walk Metropolis sampling [Hastings, 1970] proposes

$$\eta^* = \eta + cL\xi, \qquad \xi \sim N(0, I), \qquad LL' = \Omega,$$

where c and Ω are the proposal scale and covariance. For this symmetric proposal, acceptance probability is

$$\alpha(\eta, \eta^*) = \min \left\{ 1, \exp\left[\ell_\eta(\eta^*) - \ell_\eta(\eta) \right] \right\}.$$

Adaptation is restricted to warmup. Run lengths, seeds and archived settings are reported in Appendix D. The posterior estimate table appears only in Section 4.3.

B.3 Chain diagnostics and trace plots

Table 11 reports rank-normalised $\widehat{R}$, bulk ESS and tail ESS for the final notebook 03B run [Vehtari et al., 2021].

Retained-draw acceptance rates for chains 1–4 are 23.480%, 23.505%, 24.535% and 24.310%.

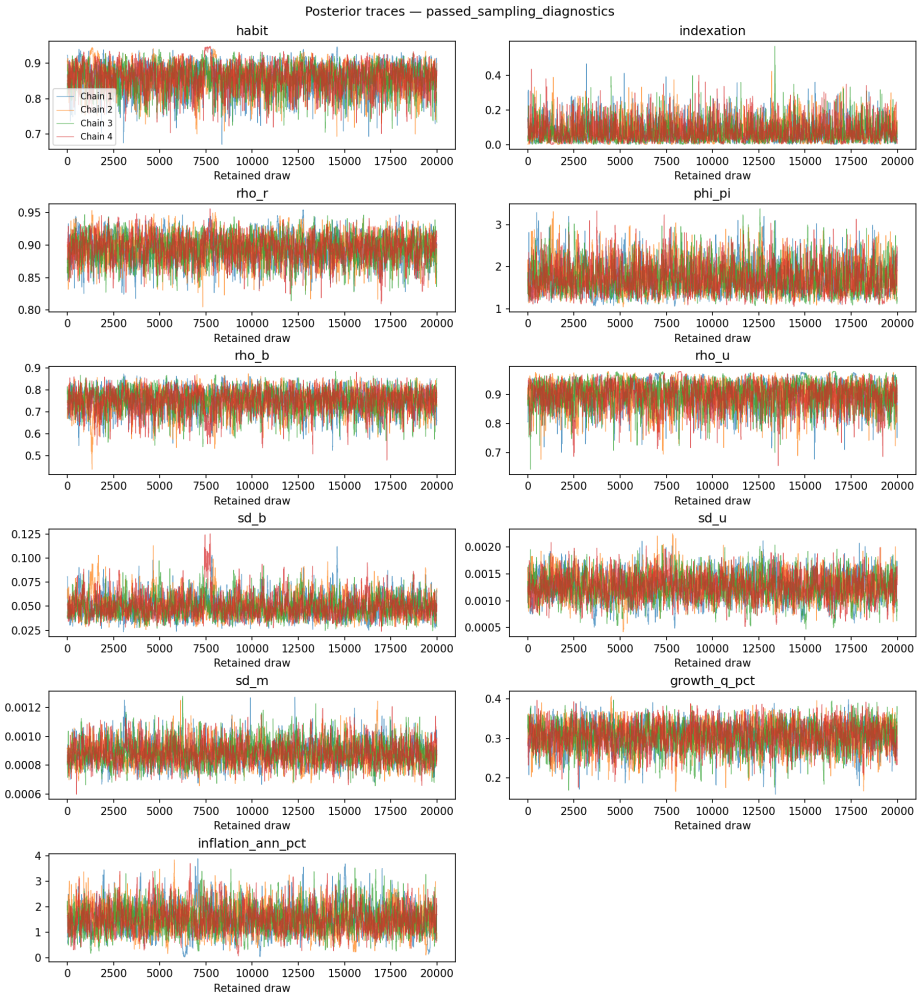


Figure 4: Posterior traces for the four final chains in notebook 03B.

Table 11: Parameter-specific sampling diagnostics

Parameter	$\widehat{R}$	Bulk ESS	Tail ESS
h_0	1.00370	1248	1006
ι	1.00397	1628	1531
ρ_r	1.00285	1608	2544
ϕ_π	1.00283	1435	2286
ρ_b	1.00254	1704	1993
ρ_u	1.00228	1103	918
s_b	1.00319	1180	1214
s_u	1.00469	1379	1506
s_m	1.00339	1715	3245
$\bar{g}$	1.00079	1706	2727
$\bar{\pi}$	1.00348	1190	1363

B.4 Numerical verification

The local moment-sensitivity calculation has a smallest-to-largest singular-value ratio of approximately 0.00204, with rank assessed using a relative threshold of 10^{-6}. It is a local numerical check, not a formal identification test. Likewise, the recovery exercise estimates one common shock-scale multiplier; it does not demonstrate joint parameter recovery.

Forecasting-stage checks cover conjugate BVAR algebra, lag ordering and recursion, matrix-normal sampling orientation, the empirical CRPS identity, filtering invariance and DSGE predictive moments. All recorded checks pass.

B.5 Conjugate BVAR computation

For the standardised VAR(2) in Section 5.1, define

$$x_t = (1, \widetilde{z}'_{t-1}, \widetilde{z}'_{t-2})'.$$

Stacking the usable training observations gives

$$Y = XB_{\text{VAR}} + E, \qquad Y \in \mathbb{R}^{82\times 3}, \quad X \in \mathbb{R}^{82\times 7}, \quad B_{\text{VAR}} \in \mathbb{R}^{7\times 3}.$$

The coefficient ordering is

$$B_{\text{VAR}} = \begin{pmatrix} c' \\ A'_1 \\ A'_2 \end{pmatrix}.$$

Under the conjugate prior [Karlsson, 2013],

$$B_{\text{VAR}} \mid \Sigma \sim MN_{7,3}(B_0, V_0, \Sigma), \qquad \Sigma \sim IW(S_0, \nu_0),$$

so column-wise vectorisation gives

$$\text{vec}(B_{\text{VAR}}) \mid \Sigma \sim N\left(\text{vec}(B_0), \Sigma \otimes V_0\right).$$

Table 12: Numerical verification

Check	Result	Scope
Independent equilibrium residual	1.38×10^{-14}	Evaluates unreduced economic equations.
Analytical restricted-model comparison	Passed	Compares solutions without habit, indexation and smoothing.
Passive policy rejection	Passed	Rejects an indeterminate restricted-model example.
Long-run shock decay	Passed	Checks convergence back to equilibrium.
Stationary covariance identity	Passed	Verifies the discrete Lyapunov equation.
Kalman versus dense Gaussian likelihood	Difference 2.22×10^{-15}	Compares likelihood calculations on a short synthetic sample.
Filtering prefix invariance	Passed	Future observations do not alter past filtered results.
Common shock-scale recovery	True 1; estimate 1.01968	Estimates a common scale using 600 simulated observations.
Local moment sensitivity	Rank 11	Checks numerical sensitivity to estimated parameters.

The prior settings are specified in Section 5.1.

The posterior quantities are

$$V_n = (V_0^{-1} + X'X)^{-1}, \tag{58}$$

$$B_n = V_n(V_0^{-1}B_0 + X'Y), \tag{59}$$

$$\nu_n = \nu_0 + 82 = 87, \tag{60}$$

$$S_n = S_0 + (Y - XB_n)'(Y - XB_n) \tag{61}$$

$$+ (B_n - B_0)'V_0^{-1}(B_n - B_0). \tag{62}$$

Consequently,

$$\Sigma \mid Y \sim IW(S_n, \nu_n), \qquad B_{\text{VAR}} \mid \Sigma, Y \sim MN_{7,3}(B_n, V_n, \Sigma).$$

For each independent draw, sample $\Sigma^{(r)}$ from its inverse-Wishart posterior and then set

$$B_{\text{VAR}}^{(r)} = B_n + L_V Z^{(r)} L_\Sigma',$$

where

$$L_V L_V' = V_n, \qquad L_\Sigma L_\Sigma' = \Sigma^{(r)},$$

and the 7×3 entries of $Z^{(r)}$ are independent standard normal variables. This produces the required covariance $\Sigma^{(r)} \otimes V_n$ without MCMC.

C Additional Empirical Results

C.1 Complete forecast evaluation

Table 13 supplements the main results with MAE, bias and complete subperiod metrics. Subperiod membership is determined by the target quarter.

Table 13: Complete forecast metrics

Model	Var.	h	n	RMSE	MAE	Bias	Cov.	W	IS	CRPS
Full evaluation sample: 2020–2025										
DSGE	G	1	24	4.346	1.799	0.239	79.2	1.880	30.030	1.699
BVAR	G	1	24	4.055	1.688	-0.021	79.2	2.024	25.610	1.531
BM	G	1	24	3.415	1.433	0.090	–	–	–	–
DSGE	P	1	24	2.158	1.807	-0.496	70.8	4.377	10.626	1.301
BVAR	P	1	24	2.674	2.021	-1.148	75.0	4.745	15.320	1.517
BM	P	1	24	2.191	1.752	-0.025	–	–	–	–
DSGE	R	1	24	0.551	0.427	0.113	70.8	1.182	2.662	0.314
BVAR	R	1	24	0.948	0.631	-0.061	54.2	1.101	5.157	0.488
BM	R	1	24	0.443	0.287	-0.102	–	–	–	–
DSGE	G	4	21	0.358	0.307	0.254	100.0	2.327	2.327	0.233
BVAR	G	4	21	0.551	0.320	-0.203	95.2	2.026	2.692	0.263
BM	G	4	21	0.577	0.383	-0.147	–	–	–	–
DSGE	P	4	21	3.341	2.207	-1.629	71.4	6.098	23.172	1.833
BVAR	P	4	21	3.914	2.645	-2.436	71.4	4.825	30.393	2.228
BM	P	4	21	4.362	3.455	-0.395	–	–	–	–
DSGE	R	4	21	1.426	1.038	0.217	81.0	2.807	7.125	0.772
BVAR	R	4	21	2.439	1.765	-0.582	61.9	3.304	14.167	1.349
BM	R	4	21	1.648	1.192	-0.490	–	–	–	–
Target quarters in 2020–2021										
DSGE	G	1	8	7.517	4.944	0.441	37.5	1.949	86.399	4.727
BVAR	G	1	8	7.013	4.597	0.105	37.5	2.505	73.261	4.235
BM	G	1	8	5.900	3.809	0.150	–	–	–	–
DSGE	P	1	8	2.375	2.223	-0.492	62.5	4.409	10.677	1.565
BVAR	P	1	8	2.607	2.329	-0.850	75.0	5.899	11.207	1.584
BM	P	1	8	2.335	2.038	-0.685	–	–	–	–
DSGE	R	1	8	0.647	0.524	0.074	50.0	1.171	3.175	0.381
BVAR	R	1	8	1.504	1.112	-0.045	37.5	1.379	11.235	0.900
BM	R	1	8	0.074	0.047	0.020	–	–	–	–
DSGE	G	4	5	0.333	0.302	0.089	100.0	2.346	2.346	0.227
BVAR	G	4	5	1.045	0.749	-0.688	80.0	2.298	5.093	0.547
BM	G	4	5	1.060	0.824	-0.806	–	–	–	–
DSGE	P	4	5	2.793	2.288	-1.835	60.0	6.224	16.023	1.684
BVAR	P	4	5	3.391	2.944	-2.605	60.0	5.398	23.924	2.222
BM	P	4	5	4.381	3.998	-3.659	–	–	–	–
DSGE	R	4	5	2.411	1.854	0.026	60.0	2.978	19.556	1.489
BVAR	R	4	5	4.067	3.086	-0.536	40.0	4.243	33.235	2.527
BM	R	4	5	0.140	0.123	0.123	–	–	–	–
Target quarters in 2022–2025										
DSGE	G	1	16	0.281	0.227	0.138	100.0	1.845	1.845	0.185
BVAR	G	1	16	0.278	0.233	-0.084	100.0	1.784	1.784	0.179
BM	G	1	16	0.293	0.245	0.060	–	–	–	–
DSGE	P	1	16	2.040	1.599	-0.499	75.0	4.361	10.601	1.168
BVAR	P	1	16	2.707	1.868	-1.296	75.0	4.168	17.377	1.484
BM	P	1	16	2.115	1.610	0.305	–	–	–	–
DSGE	R	1	16	0.496	0.378	0.133	81.2	1.187	2.406	0.280
BVAR	R	1	16	0.465	0.391	-0.070	62.5	0.962	2.118	0.282
BM	R	1	16	0.541	0.406	-0.163	–	–	–	–
DSGE	G	4	16	0.366	0.309	0.306	100.0	2.321	2.321	0.234
BVAR	G	4	16	0.239	0.186	-0.051	100.0	1.941	1.941	0.174
BM	G	4	16	0.293	0.245	0.060	–	–	–	–
DSGE	P	4	16	3.494	2.181	-1.564	75.0	6.059	25.406	1.879

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Table continued

Model	Var.	h	n	RMSE	MAE	Bias	Cov.	W	IS	CRPS
BVAR	P	4	16	4.064	2.552	-2.383	75.0	4.646	32.414	2.230
BM	P	4	16	4.356	3.286	0.625	–	–	–	–
DSGE	R	4	16	0.923	0.783	0.277	87.5	2.753	3.240	0.548
BVAR	R	4	16	1.624	1.352	-0.596	68.8	3.011	8.209	0.980
BM	R	4	16	1.886	1.526	-0.682	–	–	–	–

G: quarterly output growth; P: annualised quarterly inflation; R: annualised interest rate; BM: benchmark. Cov.: coverage in percent; W: mean interval width; IS: mean interval score. Intervals have nominal coverage of 90%. Errors are forecast minus observation. All metrics except coverage use the original percentage or percentage-point scale. Benchmark predictive distributions were not constructed.

C.2 Sensitivity to fixed calibration

All eleven estimated parameters are reoptimised under each calibration variant, using the same estimated-parameter priors and two starting points. These are MAP sensitivity exercises, not separate posterior estimations or marginal-likelihood comparisons. The baseline is not replaced by a calibration selected using test-period performance.

Table 14: Calibration sensitivity at reoptimised MAP estimates

Scenario	Log likelihood	$\bar{g}$	$\bar{\pi}$	$\bar{i}$	Rate gap	Error ACF(1)
Baseline	-233.348	0.316	1.379	3.042	1.294	0.425
$\beta = 0.995$	-248.000	0.309	1.071	4.310	2.562	0.422
$\beta = 0.9999$	-230.521	0.318	1.480	2.791	1.044	0.427
$\sigma = 1.5$	-235.875	0.271	1.755	3.781	2.033	0.441
$\sigma = 2$	-234.794	0.219	2.422	4.572	2.825	0.460

The rate gap is the long-run model rate minus the observed training-sample mean, computed before rounding. Error ACF(1) refers to marginally standardised rate prediction errors at the corresponding MAP estimate. All selected optimisation results indicate success.

Complete parameter vectors and starting-point outcomes are retained in `sensitivity_MAP_parameters.csv` and the scenario-specific `sensitivity_optimization_*.csv` files.

C.3 Additional validation and forecast figures

For each of the 1,000 posterior predictive replications, a posterior parameter draw determines the stationary covariance and state-space matrices. Simulation uses

$$s_1^{(r)} \sim N(0, P_\infty^{(r)}), \qquad s_{t+1}^{(r)} = F^{(r)} s_t^{(r)} + J^{(r)} \varepsilon_{t+1}^{(r)}, \qquad z_t^{(r)} = \mu^{(r)} + C^{(r)} s_t^{(r)}.$$

Each replication contains 84 observations. For statistic $\mathcal{T}$, a lower-tail fraction is

$$\widehat{p}_{\text{lower}} = \frac{1}{1000} \sum_{r=1}^{1000} \mathbf{1}\{\mathcal{T}(z^{(r)}) \leq \mathcal{T}(z)\}.$$

This is a descriptive posterior predictive tail fraction. A value of zero means that no replication crossed the observed statistic, rather than that the model assigns it zero probability.

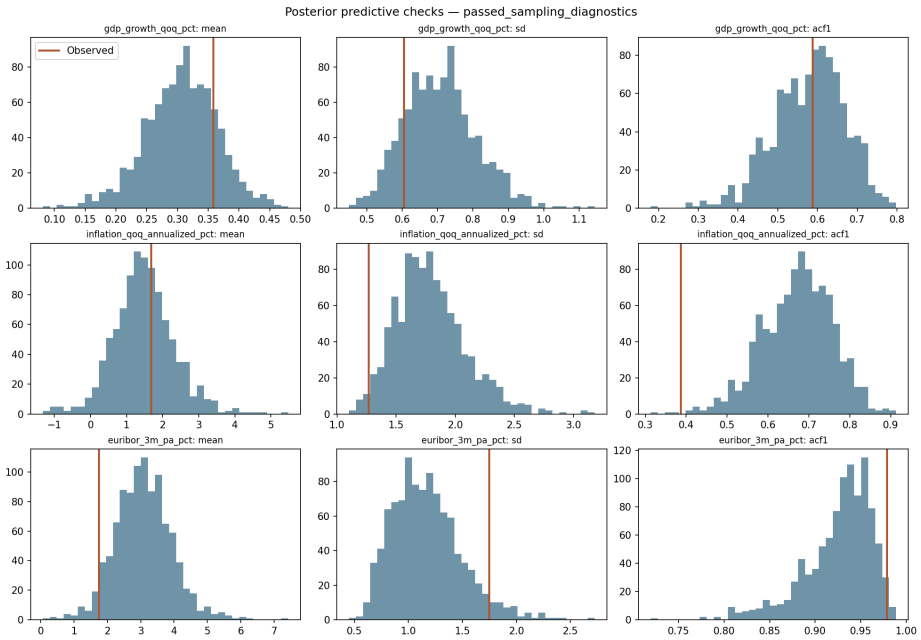


Figure 5: Posterior predictive distributions of training-sample means, standard deviations and first-order autocorrelations. Vertical lines indicate observed statistics.

The MAP filter diagnostic uses marginally standardised innovations

$$e_{j,t}^{\mathrm{std}} = \frac{v_{j,t}}{\sqrt{S_{jj,t}}}.$$

These differ from jointly whitened innovations obtained using a Cholesky factor of S_t. Although each prediction uses earlier observations for filtering, parameters are estimated on the complete training sample; the diagnostic is therefore in-sample. Detailed statistics and the in-sample prediction figure remain in the archived outputs.

At the baseline MAP estimate, the marginally standardised Euribor prediction errors have mean -0.438 and lag-one autocorrelation 0.425. Since these errors are defined as observation minus prediction, the negative mean indicates systematic overprediction, while the positive autocorrelation indicates remaining serial dependence. These are descriptive in-sample diagnostics, with parameters estimated using the complete training sample.

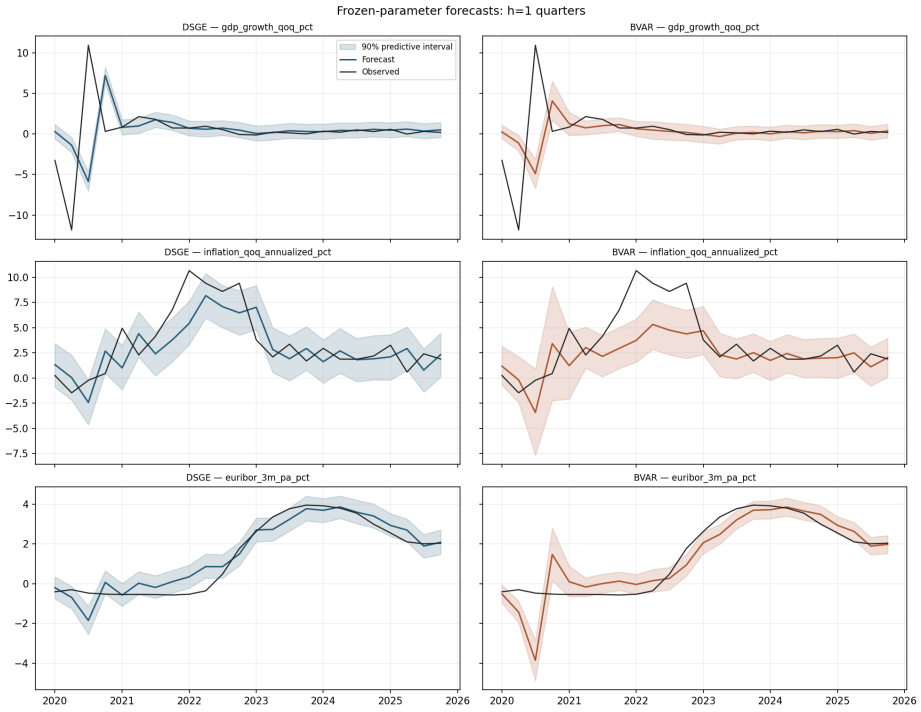


Figure 6: One-quarter-ahead forecasts and pointwise 90% predictive intervals. Parameters and posterior weights remain fixed after 2019Q4, while states and observed lags are updated at each origin.

C.4 Forecast simulation and evaluation metric definitions

Predictive simulation. For DSGE draw r and forecast origin o, filtering the observation prefix gives

$$s_o^{(r)} \sim N\left(a_{o|o}^{(r)}, P_{o|o}^{(r)}\right).$$

Future states and observations are simulated using

$$s_{o+j}^{(r)} = F^{(r)} s_{o+j-1}^{(r)} + J^{(r)} \varepsilon_{o+j}^{(r)}, \qquad z_{o+j}^{(r)} = \mu^{(r)} + C^{(r)} s_{o+j}^{(r)}.$$

The draw-specific conditional mean at horizon h is

$$m_{o,h}^{(r)} = \mu^{(r)} + C^{(r)} \left(F^{(r)}\right)^h a_{o|o}^{(r)}.$$

Point forecasts average these means using the unchanged training-posterior weights.

BVAR predictive simulation starts from the two latest observed standardised lags and recursively applies

$$\widetilde{z}_{o+j}^{(r)} = c^{(r)} + A_1^{(r)} \widetilde{z}_{o+j-1}^{(r)} + A_2^{(r)} \widetilde{z}_{o+j-2}^{(r)} + \epsilon_{o+j}^{(r)}, \qquad \epsilon_{o+j}^{(r)} \sim N(0, \Sigma^{(r)}).$$

Results are transformed back using the fixed training means and standard deviations. Point forecasts average recursions with future innovations set to zero. Predictive intervals use the 5th and 95th percentiles of simulated future observations.

Point forecast metrics. For one variable, model, horizon and target sample, let y_j be the realised observation and $\widehat{y}_j$ the point forecast. With $e_j = \widehat{y}_j - y_j$,

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n} e_j^2}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n} |e_j|, \qquad \mathrm{Bias} = \frac{1}{n}\sum_{j=1}^{n} e_j.$$

Positive bias indicates overprediction.

Interval metrics. For central $(1-\alpha)$ intervals $[L_j, U_j]$, with $\alpha = 0.1$,

$$\mathrm{Coverage} = \frac{100}{n}\sum_{j=1}^{n} \mathbf{1}\{L_j \le y_j \le U_j\}, \qquad \mathrm{Width} = \frac{1}{n}\sum_{j=1}^{n} (U_j - L_j).$$

The interval score is

$$\mathrm{IS}_\alpha(L_j, U_j; y_j) = (U_j - L_j) + \frac{2}{\alpha}(L_j - y_j)\mathbf{1}\{y_j < L_j\} + \frac{2}{\alpha}(y_j - U_j)\mathbf{1}\{y_j > U_j\}.$$

Reported scores are averages across targets. The score penalises both width and observations outside the interval [Gneiting and Raftery, 2007].

Continuous ranked probability score. For predictive CDF $\mathcal{F}_j$,

$$\mathrm{CRPS}(\mathcal{F}_j, y_j) = \int_{-\infty}^{\infty} \left[\mathcal{F}_j(x) - \mathbf{1}\{y_j \le x\}\right]^2 dx.$$

With R predictive draws $x_{j,r}$, the empirical-distribution score is

$$\widehat{\mathrm{CRPS}}_j = \frac{1}{R}\sum_{r=1}^{R} |x_{j,r} - y_j| - \frac{1}{2R^2}\sum_{r=1}^{R}\sum_{s=1}^{R} |x_{j,r} - x_{j,s}|.$$

The implementation evaluates this expression using sorted draws. It is the CRPS of the empirical predictive distribution, without a finite-simulation fairness adjustment. Scores are computed separately for each variable and averaged across targets; lower values are better [Gneiting and Raftery, 2007].

D Reproducibility and File Guide

D.1 Notebook order and required uploads

Run each notebook from top to bottom in a fresh Colab runtime, passing the result archives between stages without editing their internal files.

Table 15: Notebook execution order and archive dependencies

Notebook	Required uploads and resulting archive
`DSGE_01_Data_Preparation_Colab.ipynb`	Upload `data.csv` (GDP), `data(1).csv` (HICP), and `data(2).csv` (Euribor). Output: `DSGE_Data_Results.zip`.
`DSGE_02_Model_Solution_Colab.ipynb`	Upload `DSGE_Data_Results.zip`. Output: `DSGE_Model_02_Results.zip`.
`DSGE_03_Bayesian_Estimation_Colab.ipynb`	Upload `DSGE_Model_02_Results.zip`. Output: `DSGE_Estimation_03_Results.zip`.
`DSGE_03B_Validation_Colab.ipynb`	Upload `DSGE_Estimation_03_Results.zip`. Output: `DSGE_Validation_03B_Results.zip`, containing the final posterior used in the report.
`DSGE_04_BVAR_Forecasting_Colab.ipynb`	Upload both `DSGE_Validation_03B_Results.zip` and `DSGE_Data_Results.zip`. Output: `DSGE_Forecasting_04_Results.zip`.

Raw filenames reflect download order; economic identities are verified from the series keys retained in the audit. Notebook 04 uses the final notebook 03B posterior, rather than the preliminary notebook 03 posterior.

D.2 Data processing and estimation settings

Data construction is described in Section 2. Table 16 records the principal computational settings. BVAR prior details are given in Section 5.1 and Appendix B.5.

Table 16: Principal computational settings

Component	Setting
Preliminary MCMC, notebook 03	4 chains; 3,000 warmup and 6,000 retained draws per chain
Final MCMC, notebook 03B	4 chains; 5,000 warmup and 20,000 retained draws per chain; no thinning
Final MAP optimisation	2 starting points
Posterior predictive checks	1,000 replications of 84 observations
DSGE forecasting posterior	2,000 draws; 500 selected without replacement from each chain
BVAR posterior simulation	4,000 independent conjugate posterior draws
Forecast horizons	$h = 1$ and $h = 4$, targeting individual quarterly observations
Predictive intervals	5th and 95th percentiles
Impulse responses	2,000 posterior draws; 40 horizons indexed 0–39
Parameter and posterior-weight updating	None after 2019Q4
Origin-specific updating	DSGE filtered states and BVAR observed lags

D.3 Random seeds

Table 17: Seeds used in the reported workflow

Task	Seed
Model demonstration simulation	404
Independent model-equation verification	123
Baseline MAP starting-point generation	101
Preliminary MCMC chains	2026–2029
Final MCMC chains	3030–3033
Synthetic common-scale recovery	303
Posterior predictive checks	4040
Calibration-sensitivity starting points	505
BVAR posterior draws	5050
DSGE posterior-draw selection	5051
DSGE predictive simulation	5052
BVAR predictive simulation	5053

Selected DSGE draw indices and posterior predictive sampling indices are

retained in the corresponding NPZ files.

D.4 Output files and audit trail

Stage-specific files are retained in each archive's `outputs/` directory. Identically named files should remain associated with their source archive.

Table 18: Principal archived output groups

Group	Principal files
Data and audit	`observations_estimation.csv`, `observations_test.csv`, `audit.json`, `data_audit.json`, `run_config.json`
Posterior and MAP	`posterior_draws.npz`, `posterior_summary.csv`, `map_parameters.csv`, `map_state_space.npz`, `DSGE_selected_posterior.npz`, `bvar_posterior.npz`
Diagnostics and validation	`chain_info.csv`, `diagnostics.json`, `verification.json`, `posterior_predictive_statistics.csv`, `posterior_predictive_checks.csv`, `fixed_calibration_sensitivity.csv`
Forecasts and metrics	`forecasts.csv`, `predictive_draws.npz`, `forecast_metrics.csv`
Impulse responses	`DSGE_IRF_draws.npz`, `DSGE_IRF_summary.csv`

Recorded source-file and input-archive SHA-256 hashes identify the data snapshot and upstream results used at each stage. The archived `model.py`, `estimation.py` and `forecasting.py` files document the numerical implementation.

D.5 Software environment and reproducibility limits

The recorded Colab runtime uses Python 3.13.16. The data-preparation audit records NumPy 2.1.3, pandas 2.2.3 and Matplotlib 3.10.0. SciPy is also required, and the estimation configuration records use of the Numba likelihood kernel, with a NumPy fallback available.

The metadata do not provide a complete pinned dependency environment. Seeds and archived inputs support reproduction, but numerical-library and platform differences can affect results and prevent bitwise equality. Reproduction should use the archived data snapshot and input dependencies: downloading the same ECB series again may return revised historical observations.